

Decision Tree Regression

Importing libraries

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

Import datasets

```
In [8]: df = pd.read_csv("01_dataset/Position_Salaries.csv")
X = df.iloc[:, 1:-1].values
y = df.iloc[:, -1].values
```

```
In [9]: print(X)
```

```
[[ 1]
 [ 2]
 [ 3]
 [ 4]
 [ 5]
 [ 6]
 [ 7]
 [ 8]
 [ 9]
[10]]
```

```
In [10]: print(y)
```

```
[ 45000  50000  60000  80000 110000 150000 200000 300000 500000
1000000]
```

Training the Decision Tree Regression Model on the whole dataset

```
In [12]: from sklearn.tree import DecisionTreeRegressor
regressor = DecisionTreeRegressor(random_state = 0)
regressor.fit(X, y)
```

```
Out[12]: ▼ DecisionTreeRegressor ⓘ ?
```

► Parameters

Predict a new result

```
In [13]: regressor.predict([[6.5]])
```

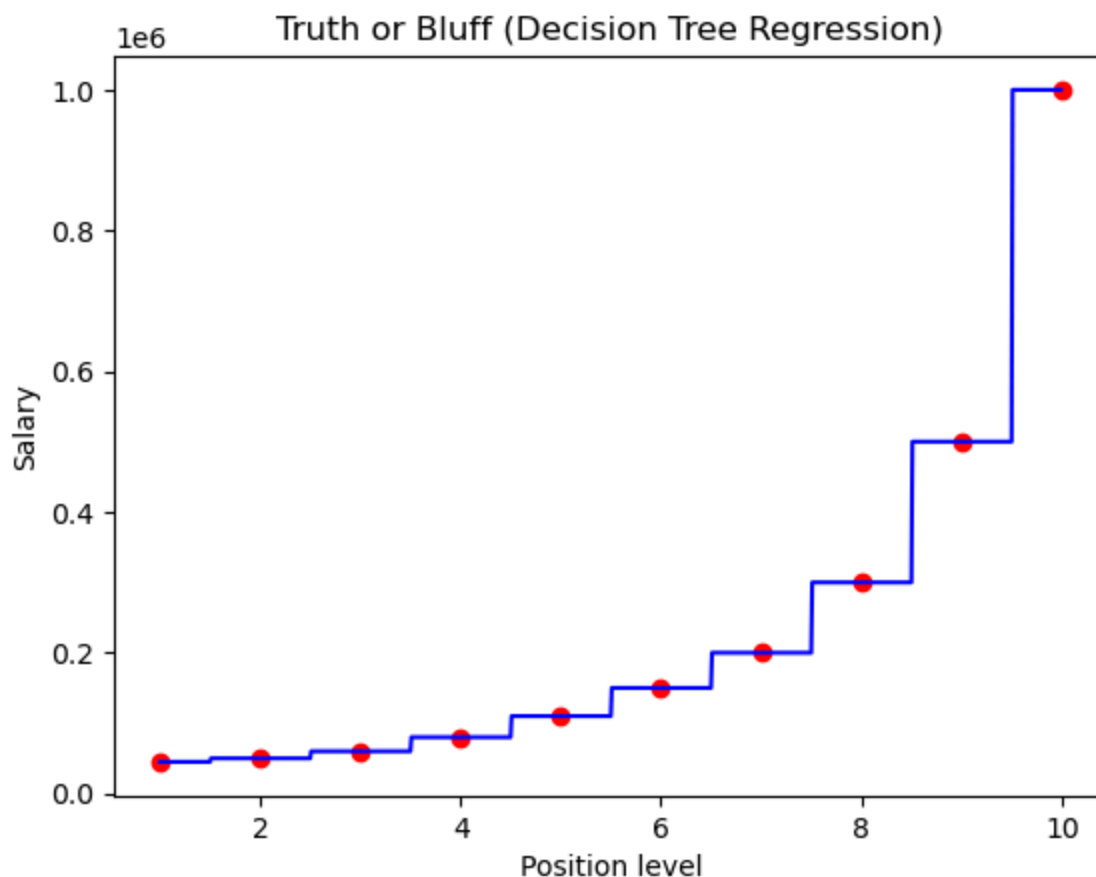
```
Out[13]: array([150000.])
```

Visualising the Decision Tree Regression results(higher resolution)

```
In [14]: X_grid = np.arange(min(X), max(X), 0.01)
X_grid = X_grid.reshape((len(X_grid), 1))
plt.scatter(X, y, color = 'red')
plt.plot(X_grid, regressor.predict(X_grid), color = 'blue')
plt.title('Truth or Bluff (Decision Tree Regression)')
plt.xlabel('Position level')
plt.ylabel('Salary')
plt.show()
```

/tmp/ipykernel_13756/1634290011.py:1: DeprecationWarning: Conversion of an array with ndim > 0 to a scalar is deprecated, and will error in future. Ensure you extract a single element from your array before performing this operation. (Deprecated NumPy 1.25.)

```
X_grid = np.arange(min(X), max(X), 0.01)
```



Note: The Decision Tree prediction model is not well adapted to 2 dimensional datasets

```
In [ ]:
```